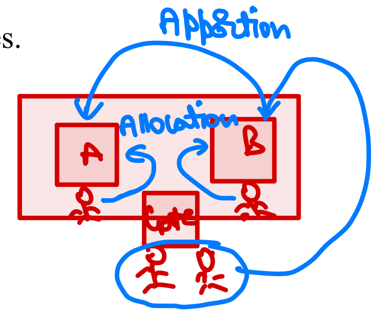
OVERHEADS

1. Overheads

It is the total of indirect material, indirect labour and indirect expenses.



2. Steps for Overheads

(A) Estimation and Distribution

(B) Recovery Rate

(C) Under or Over Recovery

3. Types of Department

(A) Production Department – Involved in manufacturing of goods or services

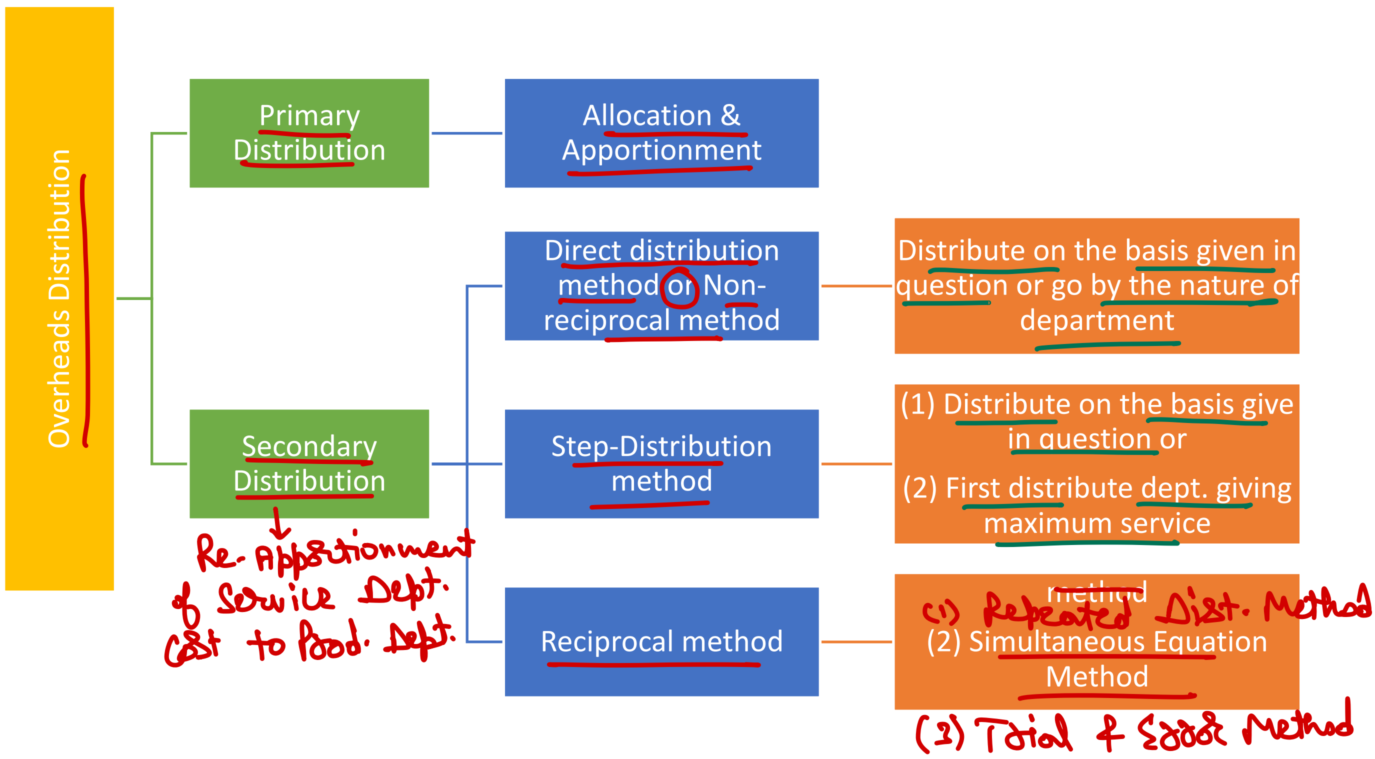
(B) Service Department – Help production department in performing their services

4. Distribution

(A) Allocation

(B) Apportionment

5. Overheads Distribution



6. Overheads Distribution Statement

Particulars	Basis	Production Department		Service Department	
		A	B	X	Y
Direct Cost	Allocation	NA	NA	✓	✓
Identified OHs	Allocation	✓	✓	✓	✓
Common OHs	Apportion	✓	✓	✓	✓
Total		✓	✓	✓	✓
Cost of Dept. X	Apportion	✓	✓	(✓)	✓
Cost of Dept. Y	Apportion	✓	✓	✓	(✓)
Total		✓	✓	Nil	Nil

7. Overheads Distribution Methods

(A) Direct Distribution Method

Expenses of service department will be distributed only to production departments irrespective of the fact whether they provide service to each other or not.

(B) Step-Distribution Method or Non-Reciprocal Method or Step-Allocation Method

- This method is to be applied only if specifically provided in question.
- Once expenses of a particular department are distributed and becomes '0' then there should not be any other amount shown in that particular department column.
- If way of distribution is provided in question then always follow that e.g. first distribute expenses of A then B, C and D.
- If no way is provided then first distribute the expenses of department which is providing maximum % of service to other service departments and so on.

(C) Reciprocal Method

(i) Repeated Distribution Method – Start with any service department and distribute in all departments. Distribute till the expenses of all service department becomes zero.

(ii) Simultaneous Equation Method – Calculate total expenses of service department using equation and then distribute to all departments.

(iii) Trial and Error Method – Use repeated distribution method only in service department to calculate its total expenses and then apportion to production department.

8. Important Points

- ➢ If no method is specified for secondary distribution then it is preferred to follow simultaneous equation method.
- ➢ Fixed expenses are apportioned on the basis of normal capacity.
- ➢ Variable expenses are apportioned on the basis of actual capacity.

Power

Pref - 1) Kwh
Pref - 2) Kw / M. hrs. / hrs.

Question – 1

SK Ltd. has three production departments P1, P2 and P3 and two service departments S1 and S2. The following data are extracted from the records of the Company for the month of October:

		₹	
Floor	Rent and rates → 20:25:30:20:5	62,500	
light Pt	General lighting → 10:15:20:10:5	7,500	
DW	Indirect Wages 3750: 2500: 3750: 1875: 625	18,750	
HP	Power 6:3:5:1:–	25,000	
Cost	Depreciation on machinery	50,000	→ 300: 400: 500 : 25 : 25
Cost	Insurance of machinery	20,000	→ 300: 400: 500 : 25 : 25

Other information:

	P1	P2	P3	S1	S2
Direct Wages (₹)	37,500	25,000	37,500	18,750	6,250
Horse power of machine used	60	30	50	10	--
Cost of machinery (₹)	3,00,000	4,00,000	5,00,000	25,000	25,000
Floor Space (Sq. Ft.)	2,000	2,500	3,000	2,000	500
Number of light points	10	15	20	10	5
Production hours worked	6,225	4,050	4,100	--	--

Expenses of the service departments, S1 and S2 are reapportioned as below:

	P1	P2	P3	S1	S2
S1	20%	30%	40%	-	10%
S2	40%	20%	30%	10%	-

Reciprocal Method

Required:

(a) Compute overhead rate per production hour of each production department

(b) Determine the total cost of product X which is processed for manufacture in department P1, P2 and P3 for 5 hours, 3 hours and 4 hours respectively, given that its direct material cost is ₹ 625 and direct labour cost is ₹ 375.

S1 = 38380 + (S2)(0.10)

S2 = 12338 + (S1)(0.10)

DM 625

DL 375

OHs[(5×9.75)+(3×15.78)+(4×20.52)]

Solution

Overheads Distribution Summary

Item of Cost	Basis of Apportionment	P1 (₹)	P2 (₹)	P3 (₹)	S1 (₹)	S2 (₹)
Direct Wages	Allocation	—	—	—	18,750	6,250
Rent and Rates	Floor Area (4:5:6:4:1)	12,500	15,625	18,750	12,500	3,125
General Lighting	Light Point (2:3:4:2:1)	1,250	1,875	2,500	1,250	625
Indirect Wages	Direct Wages (6:4:6:3:1)	5,625	3,750	5,625	2,812.5	937.5
Power	H.P. of Machines (6:3:5:1)	10,000	5,000	8,333	1,667	-
Dep. of Machine	Value-Machine (12:16:20:1:1)	12,000	16,000	20,000	1,000	1,000
Insurance of Machine	Value-Machine (12:16:20:1:1)	4,800	6,400	8,000	400	400
	Total	46,175	48,650	63,208	38,380	12,338
Cost of Dept. S1	Apportioned	8,003	12,004	16,006	(40,014)	4,001
Cost of Dept. S2	Apportioned	6,536	3,268	4,901	1,634	(16,339)

Item of Cost	Basis of Apportionment	P1 (₹)	P2 (₹)	P3 (₹)	S1 (₹)	S2 (₹)
Total Overheads		60,714	63,922	84,115	-	-
Prod. Hrs. Worked		6,225	4,050	4,100	-	-
Rate per hour (₹)		9.75	15.78	20.52		

Overheads of service cost centres Let S1 be the overhead of service cost centre S1 and S2 be the overheads of service cost centre S2.

S1 = 38,380 + 0.10 S2
S2 = 12,338 + 0.10 S1
Substituting the value of S2 in S1 we get
S2 = 38,380 + 0.10 (12,338 + 0.10 S1)
S1 = 38,380 + 1233.8 + 0.01 S1
0.99 S1 = 39,613.8
∴ S1 = ₹ 40,014
∴ S2 = 12,338 + 0.10 × 40,014 = ₹ 16339.4

Cost of Product X	**(₹)**
Direct Material	625.00
Direct Labour	375.00
Prime Cost	1,000.00
Production on Overheads	
P1 5 hours × ₹ 9.75 = 48.75	
P2 3 hours × ₹ 15.78 = 47.34	
P3 4 hours × ₹ 20.52 = 82.08	178.17
Factory Cost	1,178.17

Question – 2

BrightTech Appliances Ltd. is a leading manufacturer of smart home devices such as the SmartCool AC, PowerLite Inverter, and EcoDry Dryer. Amid increasing operational complexity, the company's finance controller has initiated a review of overhead allocations to gain better control over departmental efficiency and product profitability.

The company operates through four major departments:

1. Production – handles manufacturing across three plants using advanced machinery including stores management.
2. Administration – supports HR, finance, legal, and compliance.
3. Sales & Distribution – manages retail tie-ups, e-commerce, and logistics.
4. General Management – oversees strategy, innovation, and executive-level operations.

The extract from the budget for the next financial year is as follows:

Particulars	Amount (₹)	Notes
Raw Material Cost	5,75,00,000	Consumed in a ratio of 3:4:3 for the three products → Prod.
Indirect Material Cost	12,50,000	Manufacturing ₹7,20,000, Stores ₹25,000, General admin ₹3,80,000, Personnel ₹95,000, Sales ₹30,000

Salary & Wages	4,20,00,000 Bal. - 260L	Includes direct wages of ₹1,60,00,000 (temporary contract workers)
Rent & Property Tax	₹1,20,000	₹90,000 Warehouse Rent, ₹30,000 Property Tax
Depreciation on Non-Current Assets	₹25,00,000	Factory/office building ₹10,50,000, ACs ₹2,00,000, Machinery ₹12,50,000
Power & Fuel	₹4,90,000	₹4,70,000 for manufacturing, ₹20,000 for delivery vans
Machinery Insurance Premium	₹5,00,000	At 4% of WDV of machinery
Group Employee Insurance	₹3,10,000	Based on gross salary (excluding direct workers and top management)
Printing & Stationery	₹8,20,000	Manufacturing ₹21,000, Finance ₹5,50,000, Legal ₹28,000, Sales ₹2,21,000
Audit Fees	₹1,40,000	Statutory and internal audit
Electricity Expenses	₹4,00,000	Metered use: Production 5,600, Admin 10,000, Sales & Distribution 4,000, General Management 1,600 units
Telephone & Mobile	₹4,95,000	Production ₹1,25,000, Personnel ₹50,000, General Mgmt ₹30,000, Sales ₹75,000, Customer support ₹2,15,000
Meal Coupon Subsidy	₹2,25,000	₹3,000 per employee on roll
Software License renewals	₹16,50,000	Sores Management ₹8,50,000, Finance ₹1,50,000, Stores ₹30,000, Customer support ₹6,20,000
Misc. Expenditure	₹9,50,000	Allocated based on direct expenses

Additional Information:

- Employee Count: Production 22, Admin 20, Sales & Distribution 28
- Average Gross Salary: Production ₹6,30,000, Admin ₹4,20,000, Sales & Distribution ₹3,50,000, General Mgmt ₹2,80,000
- Floor Area: Production 9,900 m², Admin 3,600 m², Sales & Distribution 2,700 m², General Mgmt 1,800 m²
- AC Tonnage: Production 6,000 RT, Admin 3,000 RT, S&D 3,000 RT, General Mgmt 1,500 RT
- Depreciation Rates: Building 5%, AC 15%, Machinery 10%
- Department Roles:
  - o General Mgmt: 70% Sales strategy, 20% Production/Marketing, 10% Admin
  - o Admin: 50% Production, 50% Sales.

Required:

(i) Prepare a schedule of Cost Allocation for Production Dept., Administration Dept., Sales & Distribution Dept. and General Management.

(ii) Prepare a schedule of Cost Apportionment (Primary Distribution) for Production Dept., Administration Dept., Sales & Distribution Dept. and General Management.

(iii) Prepare a schedule of Secondary distribution of Administration Dept. and General Management costs.

Solution

(i) Schedule of Cost Allocation among the Departments

	Total	Production Dept.	Administration Dept.	Selling & Distribution Dept.	General Management
Raw Material Cost	5,75,00,000	5,75,00,000	-	-	-
Indirect Material Cost:	12,50,000	7,45,000	4,75,000	30,000	-
Salary & Wages	1,60,00,000	1,60,00,000	-	-	-
Rent & Property tax	90,000	-	-	90,000	-
Depreciation on Machinery	12,50,000	12,50,000	-	-	-
Power & fuel	4,90,000	4,70,000	-	20,000	-
Insurance premium on machinery	5,00,000	5,00,000	-	-	-
Printing & Stationery:	8,20,000	21,000	5,78,000	2,21,000	-
Audit fees	1,40,000	-	-	-	1,40,000
Telephone & Mobile expenses:	4,95,000	1,25,000	50,000	2,90,000	30,000
Travelling expenses	24,00,000	5,50,000	-	6,50,000	12,00,000
Software license renewal fees:	16,50,000	8,80,000	1,50,000	6,20,000	-
Total Allocated Direct Expenses	8,25,85,000	7,80,41,000	12,53,000	19,21,000	13,70,000

(ii) Schedule of Cost Apportionment (Primary Distribution)

	Basis	Total	Production Dept.	Administration Dept.	Selling & Distribution Dept.	General Management
Allocated Cost	Direct	8,25,85,000	7,80,41,000	12,53,000	19,21,000	13,70,000
Salary & Wages	Gross Salary (9:6:5:4)	2,60,00,000	97,50,000	65,00,000	54,16,667	43,33,333
Rent & Property tax	Floor Area (11:4:3:2)	30,000	16,500	6,000	4,500	3,000
Depreciation						
– Building	Floor Area (11:4:3:2)	10,50,000	5,77,500	2,10,000	1,57,500	1,05,000
– AC	RT (4:2:2:1)	2,00,000	88,889	44,444	44,444	22,222

Group Employee Insurance	Gross Salary (9:6:5:4)	3,10,000	1,16,250	77,500	64,583	51,667
Electricity Expenses	Units (14:25:10:4)	4,00,000	1,05,660	1,88,679	75,472	30,189
Meal Coupon Subsidy	No. of Employees (22,20,28)	2,25,000	66,000	60,000	84,000	15,000 (b.f.)
Miscellaneous Expenditure	Direct Expenses (Allocated Expenses)	9,50,000	8,97,729	14,414	22,098	15,760
Total		11,17,50,000	8,96,59,528	83,54,037	77,90,264	59,46,170

(iii) Schedule of Secondary Distribution:

	Basis	Total	Production Dept.	Administration Dept.	Selling & Distribution Dept.	General Management
Total Allocated and Apportioned costs		11,17,50,000	8,96,59,528	83,54,037	77,90,264	59,46,170
General Management	(2:1:7)	59,46,170	11,89,234	5,94,617	41,62,319	(59,46,170)
			9,08,48,762	89,48,654	1,19,52,583	
Administration dept.	(1:1)	89,48,654	44,74,327	(89,48,654)	44,74,327	
Total			9,53,23,089	-	1,64,26,910	

Question – 3	

M/s NOP Limited has its own power plant and generates its own power. Information regarding power requirements and power used are as follows:

	Production Dept. A	Production Dept. B	Service Dept. X	Service Dept. Y
	(Horse power hours)			
Needed capacity production	20,000	25,000	15,000	10,000
Used during the month of May	16,000	20,000	12,000	8,000

Needed Cap.

F = 4.20 l

V = 8.4 l

During the quarter ended September 2018, costs for generating power amounted to ₹ 12.60 lakhs out of which ₹ 4.20 lakhs was considered as fixed cost.

Actual Cap.

Service Dept. X renders service to A, B and Y in the ratio of 6:4:2 whereas department Y renders service to A and B in the ratio 4:1. The direct labour hours of Department A and B are 67,500 hours and 48,750 hours respectively. Required:

1) Prepare overheads distribution sheet
2) Calculate factory overhead per labour hour for the department A and B

Solution **Overheads Distribution Sheet**

Particulars	Basis	Total Amount	Production Department		Service Department	
			A	B	X	Y
Fixed Overheads	Needed cap. (20:25:15:10)	4,20,000	1,20,000	1,50,000	90,000	60,000
Variable Overheads	Used capacity (16:20:12:8)	8,40,000	2,40,000	3,00,000	1,80,000	1,20,000
Total		12,60,000	3,60,000	4,50,000	2,70,000	1,80,000
Cost of Dept. X	6 : 4 : 2		1,35,000	90,000	- (2,70,000)	45,000
Cost of Dept. Y	4: 1		1,80,000	45,000	-	(2,25,000)
Total			6,75,000	5,85,000	-	-
Labour hours			67,500	48,750		
Fact. OH per hr.			₹ 10	₹ 12		

Question – 4

SNS Trading Company has three Main Departments and two Service Departments. The data for each department is given below:

Departments	Expenses (₹)	Area (in Sq. Mtr.)	Number of employees
Main Department:		%	%
Purchase Department	5,00,000	12 30%	800 23.53%
Packing Department	8,00,000	15 37.5%	1700 50%
Distribution Department	3,50,000	7 17.5%	700 20.59%
Service Department:			
Maintenance Department	6,40,000	~~4~~	200 5.88%
Personnel Department	3,20,000	6 15% 40	~~250~~ 3400

The cost of Maintenance Department and Personnel Department is distributed on the basis of 'Area in Square Meters' and 'Number of Employees' respectively:

You are required to:

(i) Prepare a statement showing the distribution of expenses of service departments to the main departments using the "Step Ladder Method" of overhead distribution.

(ii) Compute the rate per hour of each Main Department, given that, the Purchase Department, Packing Department and Distribution Department works for 12 hours a day, 24 hours a day and 8 hours a day respectively. Assume that there are 365 days in a year and there are no holidays.

Solution

(i) & (ii) **Overheads Distribution Sheet**

Particulars	Basis	Main Department			Service Department	
		Purchase	Packing	Distribution	Maintenance	Personnel
Expenses	Allocation	5,00,000	8,00,000	3,50,000	6,40,000	3,20,000
Maintenance Department Expenses	Area (12:15:7:6)	1,92,000	2,40,000	1,12,000	(6,40,000)	96,000
Personnel Department Expenses	No. of Ees (8:17:7)	1,04,000	2,21,000	91,000	-	(4,16,000)
Total		7,96,000	12,61,000	5,53,000	-	-
Total Hours		12 × 365 = 4,380	24 × 365 = 8,760	8 × 365 = 2,920	-	-
Rate per hour		181.74	143.95	189.38	-	-

Working Note - 1

	Main Department			Service Department	
	Purchase	Packing	Distribution	Maintenance	Personnel
Area (in sq. mtr.)	12	15	7	-	6
% of service rendered by Maintenance Department	30%	37.50%	17.50%	-	15%
Number of Employees	800	1700	700	200	-
% of service rendered by Personnel department	23.53%	50%	20.59%	5.88%	

The usual method used for ranking the support departments for Step Down Allocation Method is % of Service rendered by one Service Department to another. Based on this, Maintenance Department provides 15% (highest %) of service to Personnel Department. Thus, first maintenance department expenses should be distributed first.

Question – 5

The following information are available for the three machines of a manufacturing department of KBC Ltd.:

	Preliminary estimates of expenses (per annum)			
	Total	**Machines**		
	(₹)	**P (₹)**	**Q (₹)**	**R (₹)**
Depreciation →	20,000	7,500	7,500	5,000
Spare parts →	10,000	4,000	4,000	2,000
Power 3:2:3 →	40,000			
Consumable stores →	10,000	~~4,000~~ 3600	~~3,000~~ 2700	~~3,000~~ 2700
Insurance of machinery Dep. – 75:75:50 →	8,000			
Indirect labour	20,000 + 20% = 24000 in 10:15:15 (DL Hrs)			
Building maintenance expenses Floor →	20,000 → 4:4:2			
Annual interest on capital outlay (×)	60,000	25,000	25,000	10,000
Monthly charge for rent and rates → Floor	10,000 → 4:4:2			
Salary of foreman (per month) → 1:1:1	20,000 × 12 = 2.40 L			
Salary of Attendant (per month) → 1:1:1	5,000 × 12 = 0.60 L			

(The foreman and the attendant control all the three machines and spend equal time on them).

The following additional information is also available:

	Machines		
	P	**Q**	**R**
Estimated Direct Labour Hours →	1,00,000	1,50,000	1,50,000
Ratio of K.W. Rating →	3	2	3
Floor Space (sq. ft.) →	40,000	40,000	20,000

365
(52)
(14)
(50)
249
× 8 hrs.
1992
(+) (50×4) = 200
2192

There are 14 holidays besides Sundays in the year, of which two were on Saturdays. The manufacturing department works 8 hours in a day but Saturdays are half days. All machines work at 85% capacity throughout the year and 2% is reasonable for breakdown. (2192 × 85%) – 2% = $\boxed{1826}$

You are required to:

Calculate predetermined machine hour rate for the above machines after taking into consideration the following factors:

- An increase of 15% in the price of spare parts
- An increase of 25% in the consumption of spare parts for machine 'Q' and 'R' only.
- 20% general increase in wage rates
- An 10% decrease in the consumption of consumable stores.

	4000	4000	2000
(+) 15%	600	600	300
	4600	4600	2300
(+) 25%	–	1150	575
	4600	5750	2875

Solution

Statement of Machine Hour Rate

Particulars	Basis	P (₹)	Q (₹)	R (₹)
Standing Charges				
Insurance	Depreciation basis	3,000	3,000	2,000
Indirect labour	Direct labour	6,000	9,000	9,000

(20,000 + 20% = 24,000)				
Building Maintenance	Floor Space	8,000	8,000	4,000
Rent & Rates	Floor Space	48,000	48,000	24,000
Salary of Foreman	Equal	80,000	80,000	80,000
Salary of attendant	Equal	20,000	20,000	20,000
Total Standing charges (A)		**1,65,000**	**1,68,000**	**1,39,000**
Machine Expenses				
Depreciation	Direct	7,500	7,500	5,000
Spare parts	Final estimates	4,600	5,750	2,875
Power	KW Rating	15,000	10,000	15,000
Consumable stores (10,000 – 10% = 9,000)	Direct	3,600	2,700	2,700
Total Machine Expenses (B)		**30,700**	**25,950**	**25,575**
Total Expenses (A + B = C)		1,95,700	1,93,950	1,64,575
Effective Machine hours (D)		1,826	1,826	1,826
Machine Hr. Rate (C÷D)		107.17	106.22	90.13

Working note – 1

Number of days in a year	=	365
Less: Sunday	=	52
Less: Holidays	=	14
Less: Saturday (52 – 2)	=	50
Full working days	=	249
Hours on full working days (249 × 8)	=	1,992
Hours on half days (50 × 4)	=	200
Total hours	=	2,192
Effective hours at 85% level	=	1,863
Less: Normal loss @2%	=	37
Effective running hours	=	1,826

Working note – 2

Particulars	P	Q	R
Preliminary estimates of Spare parts	4,000	4,000	2,000
(+) Increase in price @15%	600	600	300
	4,600	4,600	2,300
(+) Increase in consumption@25%	-	1,150	575
Estimated Cost	4,600	5,750	2,875

Working note – 3

Interest on capital outlay is a financial matter and, therefore it has been excluded from the cost accounts.

9. Recovery Rate

Rate at which overheads are recovered/absorbed/charged

Recovery Rate = Pre-determined absorption rate = $\frac{Budgeted\ overheads}{Budgeted\ recovery\ base}$

10. Type of Recovery Rate

(A) Direct Material Cost % Method = $\frac{Budgeted\ Overheads}{Budgeted\ Material\ Cost} \times 100$

(B) Direct Labour Cost % Method = $\frac{Budgeted\ Overheads}{Budgeted\ Labour\ Cost} \times 100$

(C) Direct Prime Cost % Method = $\frac{Budgeted\ Overheads}{Budgeted\ Prime\ Cost} \times 100$

(D) Unit Cost Method = $\frac{Budgeted\ Overheads}{Budgeted\ Production\ Units}$

(E) Labour Hour Rate Method = $\frac{Budgeted\ Overheads}{Budgeted\ Labour\ Hours}$

(F) Machine Hour Rate Method = $\frac{Budgeted\ Overheads}{Budgeted\ Machine\ Hours}$

11. Machine Hour Rate

It is applied in case of capital intensive units.

All overheads are divided into Fixed/Standing Charges and Variable/Running Charges

Particulars	Amount
Fixed or Standing Expenses	
Rent	
Heat & light	
Insurance	
Wages	
Depreciation (if fixed)	
Total Fixed expenses (A)	
Variable or Running Expenses	
Power	
Consumable stores	
Depreciation (if variable)	
Repair & Maintenance	
Total Variable expenses (B)	
Total Cost (A + B)	✓
Effective Machine Hours	✓
Machine Hour Rate	✓

12. Points to Remember (PTR)

(A)

Depreciation
- If Life is given in Years – Fixed
- If Life is given in Hours - Variable

(B) Effective Machine Hours = Total Machine Hours – Idle Hours

(C) Normal Idle Time – Hours during which work is not done e.g. maintenance, setup, lunch etc.

(D) Unless otherwise provided, following points are to be assumed for setup hours:
- No electricity or power is used during set-up hours
- These hours are considered to be un-productive

Question – 6	

A manufacturing unit has purchased and installed a new machine at a cost of ₹ 24,90,000 to its fleet of 5 existing machines. The new machine has an estimated life of 12 years and is expected to realize ₹ 90,000 as scrap value at the end of its working life. F Dep. $= \left[\frac{24.90L - 0.90L}{12}\right] = 2L$

Other relevant data are as follows: Eff. M.hrs. = 2496 − 312 = 2184

(i) Budgeted working hours are 2,496 based on 8 hours per day for 312 days. Plant maintenance work is carried out on weekends when production is totally halted. The estimated maintenance hours are 416. During the production hours machine set-up and change over works are carried out. During the set-up hours no production is done. A total 312 hours are required for machine set-ups and change overs.

(ii) An estimated cost of maintenance of the machine is ₹ 2,40,000 p.a. V

V $2400 \times \left(\frac{312}{6}\right) = 124800$

(iii) The machine requires a component to be replaced every week at a cost of ₹ 2,400.

(iv) There are three operators to control the operations of all the 6 machines. Each operator is paid ₹ 30,000 per month plus 20% fringe benefits. F $\frac{30000 \times 12 \times 3}{6} = 1.80L$ Salary + FR: 1.80L × 20% = 36000

(v) Electricity: During the production hours including set-up hours, the machine consumes 60 units per hour. During the maintenance the machine consumes only 10 units per hour. Rate of electricity per unit of consumption is ₹ 6. [(2496×60×6) + (416×10×6)] = 923520

(vi) Departmental and general works overhead allocated to the operation during last year was ₹ 5,00,000. During the current year it is estimated to increase by 10%.

Required to compute the machine hour rate. $\frac{(5L + 10\%)}{6} = 91667$

Solution

Effective machine hours = 2,496 – 312 = 2,184 hours

Statement of Machine Hour Rate

Particulars	**Amount (₹)**
Fixed Expenses	
Depreciation $\left[\frac{24{,}90{,}000-90{,}000}{12}\right]$	2,00,000
Operator's salary [30,000 × 3 × 12 × (1/6)]	1,80,000
Fringe Benefits (1,80,000 × 20%)	36,000
Department & General Overheads [5,00,00 × 110% × (1/6)]	91,667
Fixed expenses (A)	5,07,667
Variable Expenses	
Maintenance	2,40,000
Replacement cost $\left(2{,}400 \times \frac{312}{6}\right)$	1,24,800
Electricity during production (2,496 ×60 × 6)	8,98,560
Electricity during maintenance (416 × 10 × 6)	24,960
Variable expenses (B)	12,88,320
Total expenses (A + B = C)	17,95,987
Effective machine hours (D)	2,184
Machine hour rate (C ÷ D)	822.34

Question – 7

Calculate Machine Hour Rate from the following particulars:

Cost of Machine	-	₹ 25,00,000
Salvage Value	-	₹ 1,25,000
Estimated life of the machine	-	25,000 Hours
Working Hours (per annum)	-	3,000 Hours
Hours required for maintenance	-	400 Hours
Setting-up time required	-	8% of actual working hours

→ Dep. = $\left(\frac{25L - 1.25L}{25000}\right) \times 2407$ = ✓

Let Eff. M. Hrs. = y

3000 − 400 − (0.08)(y) = y

2600 = (1.08)(y)

$y = \frac{2600}{1.08} = 2407$

Additional Information:

(a) Power 25 units @ ₹ 5 per unit per hour. → 25 × 5 × 2407 = ✓

(b) Cost of repairs and maintenance ₹ 26,000 per annum. ✓

(c) Chemicals required for operating the machine ₹ 2,600 per month. × 12 = 31200

(d) Overheads chargeable to the machine ₹ 18,000 per month. × 12 = 2.16L

(e) Insurance Premium (per annum) 2% of the cost of machine × 25L = 0000

(f) No. of operators - 02 (looking after three other machines also) → $\frac{18500 \times 12 \times 2}{4}$ = 111000

(g) Salary per operator per month ₹ 18,500

Solution

Let effective machine hours = y

∴ set-up time = (0.08)y

Thus, y = 3,000 – 400 – (0.08)y

y = 2,407

Statement of Machine Hour Rate

Particulars	Amount (₹)
Fixed Expenses	
Chemicals (2,600 × 12)	31,200
Overheads (18,000 × 12)	2,16,000
Insurance (25,00,000 × 2%)	50,000
Salary $\left(\frac{18,500 \times 12 \times 2}{4}\right)$	1,11,000
Fixed expenses	4,08,200
Effective machine hours	2,407
Fixed expenses per machine hour	169.59
Variable Expenses per machine hour	
Depreciation $\left(\frac{25,00,000-1,25,000}{25,000}\right)$ × 2407	95
Repair & Maintenance (26,000 ÷ ~~2,407)~~ ₹	10.80
Power (25 × 5)	125
Machine hour rate	400.39

Question – 8	

A machine shop has 8 identical machines manned by 6 operators. The machine cannot work without an operator wholly engaged on it. The original cost of all the 8 machines works out to ₹ 32,00,000. The following particulars are furnished for a six months period:

6 Month 6 Month

Normal available hours per month per operator	208	(208×6×6) − [(18+20+10)×6] = 7200
Absenteeism (without pay) hours per operator	18	
Leave (with pay) hours per operator	20	wages = [(208×6×6) − (18×6)] $\left(\frac{100}{8}\right)$ = 92250
Normal unavoidable idle time – hours per operator	10	
Average rate of wages per day of 8 hours per operator	₹ 100	
Production bonus estimated	10% on wages ×92250 = 9225	
Power consumed	₹ 40,250	
Supervision and Indirect Labour	₹ 16,500	
Lighting and Electricity	₹ 6,000	

The following particulars are given for a year:

Insurance	₹ 3,60,000	× 6/12 = 1.80 l
Sundry work Expenses	₹ 50,000	× 6/12 = 0.25 l

Management Expenses allocated ₹ 5,00,000 × 6/12 = 2.50L
Depreciation 10% on the original cost × 32L × 6/12 = 1.60L
Repairs and Maintenance (including consumables) 5% of the value of all the machines × 32L × 6/12 = 0.80L
Prepare a statement showing the comprehensive machine hour rate for the machine shop.

Solution

Effective machine hour = (208 × 6 × 6) – [(18 – 20 – 10) × 6] = 7,200

Statement of Machine Hour Rate

Particulars	Amount (₹)
Fixed Expenses	
Wages [{(208 × 6 × 6) – (18 × 6)} × (100/8)]	92,250
Bonus (92,250 × 10%)	9,225
Supervision	16,500
Lighting and electricity	6,000
Insurance [3,60,000 × (6/12)]	1,80,000
Depreciation [32,00,000 × 10% × (6/12)]	1,60,000
Sundry work expenses [50,000 × (6/12)]	25,000
Management expenses allocated [5,00,000 × (6/12)]	2,50,000
Fixed expenses	7,38,975
Effective machine hours	7,200
Fixed expenses per machine hour	102.64
Variable Expenses per machine hour	
Repair & Maintenance $\left(32,00,000 \times 5\% \times \frac{6}{12} \times \frac{1}{7200}\right)$	11.11
Power $\left(\frac{40,250}{7,200}\right)$	5.59
Machine hour rate	119.34

13. Dual Recovery Rate/ Two-tier Machine Hour Rate

It is to be used in following situation:

(A) When question mention to use

(B) Job charge is for ~~separation~~ set-up and operation separately

In this case set-up hours are considered to be productive

→ For FC per machine hour – Use total hours (Production + Set-up)

→ FC per machine hour will remain same for both i.e. operation and set-up

→ VC will be computed separately for both production and set-up.

Mach. Exp. = $\left(\frac{18000}{3}\right) + \left(19.20L \times \frac{10}{100} \times \frac{1}{12}\right) + \left(12L \times 20\% \times \frac{1}{12}\right) = 42000 \div 3500 = 12$

Question – 9	

USP Ltd. is the manufacturer of 'double grip motorcycle tyres'. In the manufacturing process, it undertakes three different jobs namely, Vulcanizing, Brushing and Striping. All of these jobs require the use of a special machine and also the aid of a robot when necessary. The robot is hired from outside and the hire charges paid for every six months is ₹ 2,70,000. An estimate of overhead expenses relating to the special machine is given below:

$2.70L \times \frac{1}{6} \times \frac{1}{2000} = 22.5$

- Rent for a quarter is ₹ 18,000
- The cost of the special machine is ₹ 19,20,000 and depreciation is charged @10% per annum on straight line basis.
- Other indirect expenses are recovered at 20% of direct wages.

The factory manager has informed that in the coming year, the total direct wages will be ₹ 12,00,000 which will be incurred evenly throughout the year.

During the first month of operation, the following details are available from the job book:

M. HRS. = 3500
Robot = 2000

Jobs	Without the aid of the robot	With the aid of the robot
Vulcanizing	500	400
Brushing	1000	400
Striping	-	1200
		2000

You are required to:

(i) Compute the Machine Hour Rate for the company as a whole for a month (A) when the robot is used and (B) when the robot is not used.

(ii) Compute the Machine Hour Rate for the individual jobs i.e. Vulcanizing, Brushing and Striping.

Solution

(i) Total machine hours = 500 + 1,000 + 400 + 400 + 1,200 = 3,500

Total machine hours with the use of robot = 400 + 400 + 1,200 = 2,000

Statement of Machine Hour Rate

Particulars	Amount (₹)
Rent (18,000 ÷ 3)	6,000
Depreciation [(19,20,000 × 10%) ÷ 12]	16,000
Indirect expenses [(12,00,000 × 20%) ÷ 12]	20,000
Total expenses	42,000
Total Machine Hours	3,500
Machine hours rate (42,000 ÷ 3500)	12
Add: Robot charges per machine hour [(2,70,000 ÷ 6) ÷ 2,000]	22.50
Machine hour rate with robot charges	34.50

(ii) **Computation of Machine Hour Rate for the individual jobs**

Particulars	Vulcanizing	Brushing	Striping
OHs without robot	500 × 12 = 6,000	1,000 × 12 = 12,000	-
OHs with robot	400 × 34.50 = 13,800	400 × 34.50 = 13,800	1,200 × 34.50 = 41,400
Total	19,800	25,800	41,400

Machine hours	900	1,400	1,200
Machine hour rate	22	18.43	34.50

14. Type of Recovery Rate

(A) Departmental recovery rate = $\frac{Overheads\ of\ department}{Base\ value\ of\ department}$

(B) Blanket or plant-wise recovery rate = $\frac{Overheads\ of\ Factory}{Base\ value\ of\ Factory}$

Question – 10	

SE Limited manufactures two products – A and B. The company had budgeted factory overheads amounting to ₹36,72,000 and budgeted direct labour hour of 1,80,000 hours. The company uses pre-determined overhead recovery rate for product costing purposes. Blanket Rate = $\frac{36.72L}{1.80L}$ = 20.4

The department-wise break-up of the overheads and direct labour hours were as follows:

Particulars	Budgeted Overheads	Budgeted Direct Labour Hours	Rate per Direct Labour Hour
Department Pie	₹25,92,000	90,000 hours	Dept. Rate ₹28.80
Department Qui	₹10,80,000	90,000 hours	₹12.00
Total	₹36,72,000	1,80,000 hours	

P ⇒ (4)(A) + (1)(B) = 90000
Q ⇒ (1)(A) + (4)(B) = 90000 → A = , B =

Additional information:
Each unit of product A requires 4 hours in department Pie and 1 hour in department Qui. Also, each unit of product B requires 1 hours in department Pie and 4 hours in department Qui.

This was the first year of the company's operation. There was no WIP at the end of the year. However, 1,800 and 5,400 units of Products A and B were on hand at the end of the year.

The budgeted activity has been attained by the company. You are required to:

(i) Determine the production and sales quantities of both products 'A' and 'B' for the above year.

(ii) Ascertain the effect of using a pre-determined overhead rate instead of department-wise rates on the company's income due to its effect on stock value.

(iii) Calculate the difference in the selling price due to the use of pre-determined overhead rate instead of using department-wise overhead rates. Assume that the direct costs (material and labour costs) per unit of products A and B were ₹25 and ₹40 respectively and the selling price is fixed by adding 40% over and above these costs to cover profit and selling and administration overhead.

	A	B
	25	40
Blanket	20.4×5 = 102	20.4×5 = 102
TC	127	142
(+) 40%	50.8	
SP	177.8	198.8

	A	B
	25	40
Pie	4×28.8	1×28.80
Qie	1×12	4×12
TC	152.2	116.8
(+) 40%		
SP	213.08	163.52

Solution

(i) Computation of production and sales quantities:

The products processing times are as under –

Product	A	B	Total
Department Pie	4 hours	1 hour	90,000 hours
Department Qui	1 hour	4 hours	90,000 hours

Let X and Y be the number of units (production quantities) of the two products.

Converting these into equations, we have –

4X + Y = 90,000 &

X + 4Y = 90,000

Solving the above, we get X = 18,000; Y = 18,000

Hence, the Production and Sales Quantities are determined as under –

Product	Production Quantity	Closing Stock (Given)	Sales Quantity (Balancing Figure)
A	**18,000 units**	1,800 units	**16,200 units**
B	**18,000 units**	5,400 units	**12,600 units**

(ii) Effect of using pre-determined rate of overheads on the company's profit

Product	Closing Stock Quantity	Overhead included using pre-determined rate	Overhead included using department rate	Difference in overhead in closing stock value / Effect on closing stock value
A	1,800 units	1,800 x 5 hours x ₹ 20.40 = ₹ 1,83,600	Pie = 1,800 units x 4 hours x ₹ 28.80 = ₹ 2,07,360 Qui = 1,800 units x 1 hour x ₹ 12 = ₹ 21,600	**(-) ₹ 45,360**
B	5,400 units	5,400 x 5 hours x ₹ 20.40 = ₹ 5,50,800	Pie = 5,400 units x 1 hour x ₹ 28.80 = ₹ 1,55,520 Qui = 5,400 units x 4 hours x ₹12 = ₹ 2,59,200	**(+) ₹ 1,36,080**
Total		₹ 7,34,400	₹ 6,43,680	**(+) ₹ 90,720**

Use of pre-determined overhead rate has resulted in over valuation of stock by ₹ 90,720 due to which the company's income would be affected (increase) by ₹ 90,720. Profit would be affected only to the extent of Overhead contained in closing finished goods and closing WIP, if any.

(iii) Effect of using pre-determined on the products' selling prices

Particulars	Product A	Product B
Selling Price per unit if pre-determined overhead rate is used	₹177.80	₹ 198.80
Selling Price per unit if department wise rate is used	₹ 213.08	₹163.52
Difference	**₹ 35.28 Under-Priced**	**₹ 35.28 Over-Priced**

Workings:

(1) Pre-determined overhead recovery rate = $\frac{₹\ 36,72,000}{1,80,000 \text{ hours}}$ = ₹ 20.40 per direct labour hour

(2) If pre-determined recovery rate is used

Particulars	Product A in ₹	Product B in ₹
Materials & Labour	25.00	40.00
Add: Production Overhead	102.00	102.00
A = 5 hours x ₹ 20.40 per hour		
B = 5 hours x ₹ 20.40 per hour		
Cost of production	127.00	142.00
Add: 40% of margin	50.80	56.80
	177.80	**198.50**

(3) If department-wise recovery rate is used

Particulars	Product A in ₹	Product B in ₹
Materials & Labour	25.00	40.00
Add: Production Overhead	127.20	76.80
A = Pie = 4 hours x ₹ 28.80		
Qui = 1 hour x ₹ 12		
B = Pie = 1 hour x ₹ 28.80		
Qui = 4 hours x ₹ 12		
Cost of production	152.20	116.80
Add: 40% of margin	60.88	46.72
Selling Price per unit	**213.08**	**163.52**

15. Under or Over Recovery of Overheads

Recovered OHs = RR × Actual Base Value
Actual Overheads = data from question

- Actual OHs > Recovered OHs → Loss — Under Recovered OHs
- Actual OHs < Recovered OHs → Pft. — Over Recovered OHs

16. Treatment of Over or Under Recovery of Overheads

Treatment of Overheads

1. **Due to defective planning or factory inefficiency**
   Trf. to C.P&L
   Amount charge to P&L
   → **Under Recovery**
   Costing P&L A/c Dr.
   To OHs Control A/c
   → Over Recovery
   OHs Control A/c Dr.
   To Costing P&L A/c

2. **Due to seasonal Nature of business**
   No Entry required
   Balances are carried forward to next period

3. **Due to change in price level**
   ↳ Pass to Consumer
   Net Pft Impact
   → Under Recovery
   Sale — Cost of Sales A/c Dr. → Profit ↓
   FG St. — FG ledger control A/c Dr.
   WIP St. — WIP ledger control A/ Dr. → Profit ↑
   To OHs Control A/c

17. Supplementary Rate

$$\text{Supplementary rate} = \frac{Under/OverRecovery}{Equivalent\ units\ of\ FG}$$

Question – 11	

ABS Enterprises produces a product and adopts the policy to recover factory overheads applying blanket rate based on machine hours. The cost records of the concern reveal following information:

Budgeted production overheads	→	₹ 10,35,000
Budgeted machine hours	→	90,000
Actual machine hours worked	→	45,000

RR = 10.35L / 0.90L = 11.50

Rec. OH = 45000×11.50
= 517500

Actual production overheads ₹ 8,80,000

Production overheads (actual) include –

Paid to worker as per court's award (-) ₹ 50,000

Wages paid for strike period (-) ₹ 38,000

Stores written off (-) ₹ 22,000

Expenses of previous year booked in current year (-) ₹ 18,500

Actual OH = 751500

Production –

Finished goods 30,000 units

Sale of finished goods 27,000 units

Under Rec. = 751500 – 517500 = 2.34L

Def. Plan. = 2.34L × 1/3 = 78000 → P&L

Inr. in Cost = 2.34L × 2/3 = 1.56L → COS: 27000×5.2 = 140400; FG&CC: 3000×5.2 = 15600

The analysis of cost information reveals that 1/3 of the under absorption of overheads was due to defective production planning and the balance was attributable to increase in costs.

You are required:

(i) To find out the amount of under absorbed production overheads.

(ii) To give the ways of treating it in Cost Accounts

(iii) To apportion the under absorbed overheads over the items.

Supp. Rate = 156000 / 30000 = 5.20

WIP (40%) Comp. 5000 unit

Eq. = 30000 + (5000×40%) = 32000

Solution

(i)

		Amount (₹)
Total production overheads actually incurred during the period		8,80,000
Less: Amount paid to worker as per court order	50,000	
Less: Expenses of previous year booked in current year	18,500	
Less: Wages paid for the strike period under reward	38,000	
Less: Obsolete material written off	22,000	(1,28,500)
		7,51,500
Less: Production overheads absorbed (45,000 x ₹ 11.5)		5,17,500
Under recovered overheads		2,34,000

Budgeted machine hour rate = $\frac{10,35,000}{90,000\ hours}$ = ₹ 11.50 per hour

(ii) As one third of the under absorbed overheads i.e. ₹ 78,000 (₹ 2,34,000 × 1/3) were due to defective production policies, this being abnormal, hence should be debited to profit and loss account.

(iii) Amount of balance under absorbed overheads = ₹ 2,34,000 – 78,000 = ₹ 1,56,000

Supplementary rate = $\frac{1,56,000}{30,000\ units}$ = ₹ 5.20 per equivalent unit

	Amount (₹)
Finished stock (27,000 units × 5.20)	1,40,400
Cost of sales (3,000 units × 5.20)	15,600
Total	1,56,000

	M-1	M-2	OH
Rec. OH	30×4.5 = 135	32.5×6.5 = 211.25	2400×0.8 = 1920
Ac. OH	40×4.5 = 180	40×6.5 = 260	3000×0.8 = 2400
UR	45	48.75	480
Abs.	30	32.50	2400
Supp. Rate	1.5	1.5	0.2

Actual OH = Budget OH (Plan)

Question – 12	

SK engineering factory fabricates machine parts to customers. The factory commenced fabrication of 12 Nos. machine parts to customer's specifications and the expenditure incurred on the job for the week ending 21st August, is given below:

	(₹)	(₹)
Direct materials (all items)		78.00
Direct labour (manual) 20 hours @ ₹ 1.50 per hour		30.00
Machine facilities:		
Machine No. I : 4 hours @ ₹ 4.50	18.00	
Machine No. II: 6 hours @ ₹ 6.50	39.00	57.00
Total		165.00
Overheads @ ₹ 0.80 per hour on 20 manual hours		16.00
Total cost		181.00

The overhead rate of ₹ 0.80 per hour is based on 3,000 man hours per week; similarly, the machine hour rates are based on the normal working of Machine Nos. I and II for 40 hours out of 45 hours per week.

After the close of each week, the factory levies a supplementary rate for the recovery of full overhead expenses on the basis of actual hours worked during the week. During the week ending 21st August, the total labour hours worked was 2,400 and machine Nos. I and II had worked for 30 hours and 32½ hours respectively.

Prepare a cost sheet for the job for the fabrication of 12 Nos. machine parts duly levying the supplementary rates.

Solution

Statement of Cost

Particulars	₹	₹
Material		78
Labour 20 hours @ ₹ 1.50		30
Machine facilities:		
Machine No. I: 4 hours @ ₹ 4.50	18	
Machine No. II: 6 hours @ ₹ 6.50	39	57
Overheads 20 hours @ ₹ 0.80 per hour		16
		181
Supplementary Rates		
Overheads 20 hours @ ₹ 0.20 per hour	4	
Machine facilities:		
Machine No. I: 4 hours @ ₹ 1.50	6	
Machine No. II: 6 hours @ ₹ 1.50	9	19
Total Cost		**200**

Working notes:

Overheads budgeted: 3,000 hours @ ₹ 0.80 = ₹ 2,400

Actual hours: 2,400 hours

Actual rate per hour ₹ 2,400/2,400 hours = ₹ 1

Supplementary charge ₹ 0.20 (₹ 1 – 0.80) per hour

Machine facilities:

	Machine No. I	**Machine No. II**
Budgeted	(40 × ₹ 4.50) = ₹ 180	(40 × ₹ 6.50) = ₹ 260
Actual number of hours	30	32½
Actual rate per hour	₹ 6	₹ 8
Supplementary rate per hour	₹ 6 – ₹ 4.50 = ₹ 1.50	₹ 8 – ₹ 6.50 = ₹ 1.50